



UNIVERSITY OF CALOOCAN CITY
COMPUTER ENGINEERING DEPARTMENT



Software Design

Progress Report No. 8

Data Structures and Algorithms

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I. Objectives

In this section, the goals in this laboratory are:

- To describe a good conceptual design of the system
- To identify one of the three levels of structure of program – Data Structure

II. Methods

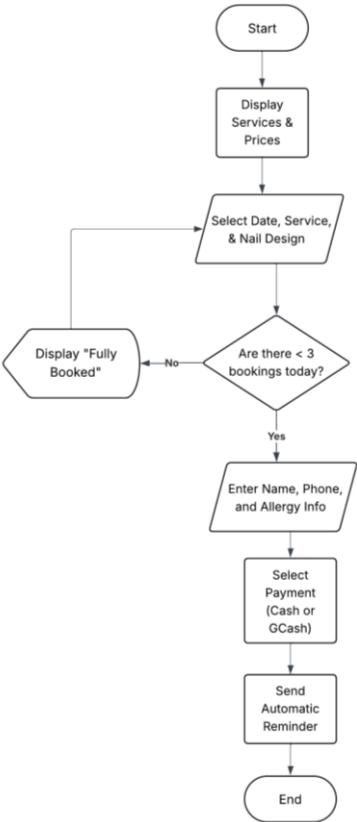
Algorithm

This algorithm outlines the logical, step-by-step process for a client to book a service, from the initial selection to the final booking confirmation.

1. **Start.**
2. **Display Services:** Show options for Gel overlay, BIAB, and Soft gel extensions.
3. **Input Selection:** Client chooses their service and optional nail art design.
4. **Check Availability:** System checks if the daily limit of three clients has been reached.
If < 3 , proceed.
Else, prompt the user to select another date.
5. **Collect Info:** Gather client name, phone number, and allergy details.
6. **Payment Processing:** Record payment type as Cash or GCash.
7. **Notification:** Send an automatic reminder to the client.
8. **End.**

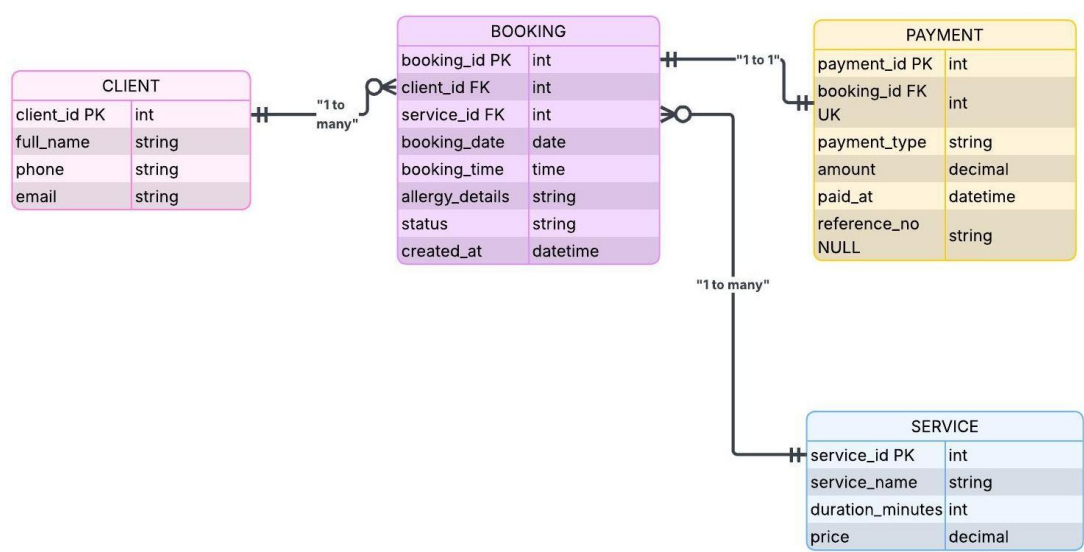
Flowchart

The flowchart illustrates the decision logic, specifically the daily booking limit and the transition to automatic reminders.



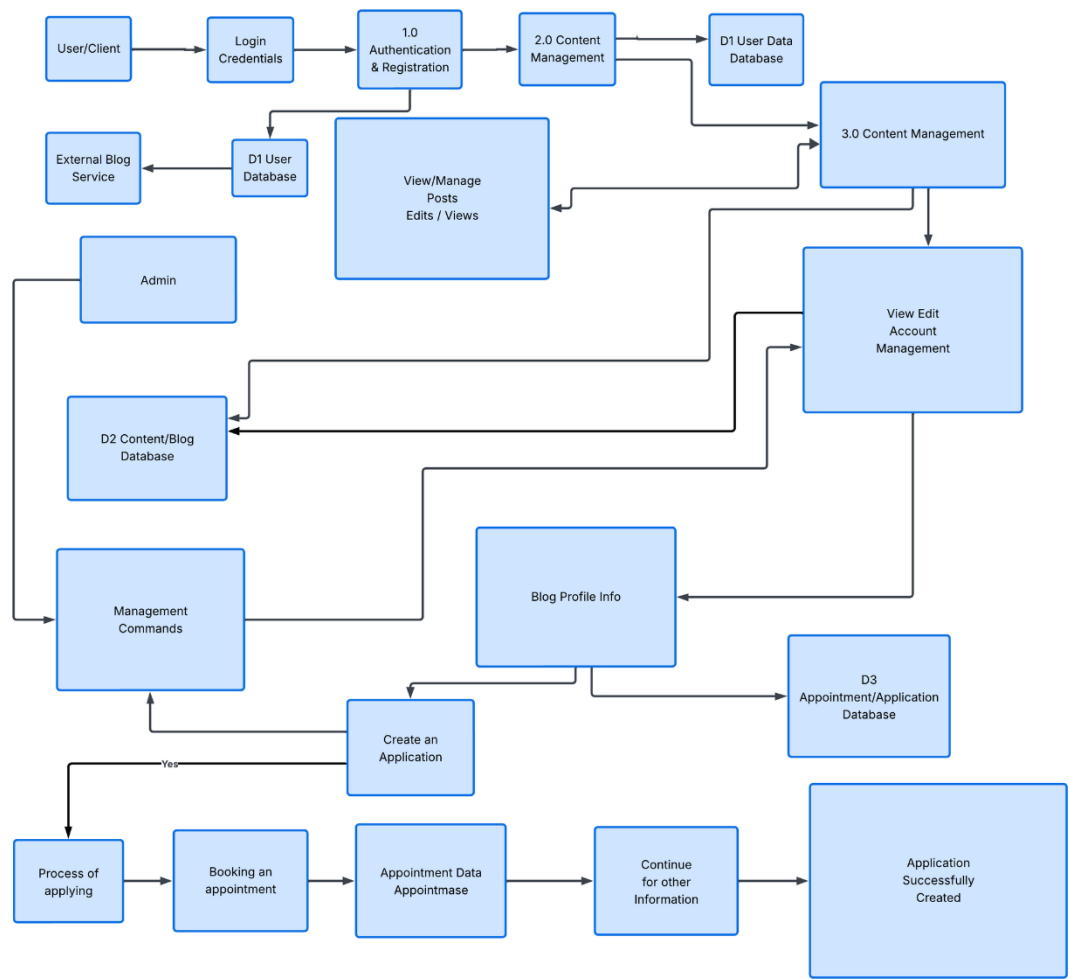
Entity-Relationship Diagram (ERD)

This diagram establishes the structural framework of the system by defining the relationships between core entities such as the Client, Booking, Payment, and Service.



Data Flow Diagram (DFD)

Illustrates the movement of information. It tracks how data flows from the initial "Client" external entity through various processes such as "Check Availability" and "Process Payment" until it results in a notification log entry.



II. Conclusion

In this report, our group successfully created a conceptual design for our nail salon booking system. We were able to identify the important data structures and logic needed to make the program work. By writing an algorithm and drawing a flowchart, we clearly defined how the system should handle client choices like picking between a Gel overlay or soft gel extensions and how it should check for the 3-client daily limit. Using the ERD and DFD helped us visualize how to store client names, phone numbers, and allergy details, as well as how to record payments through Cash or GCash. We believe this plan gives us a strong foundation for the next steps of our project. Overall, this design ensures that the booking process is organized, easy to follow, and helpful for both the salon and the clients.